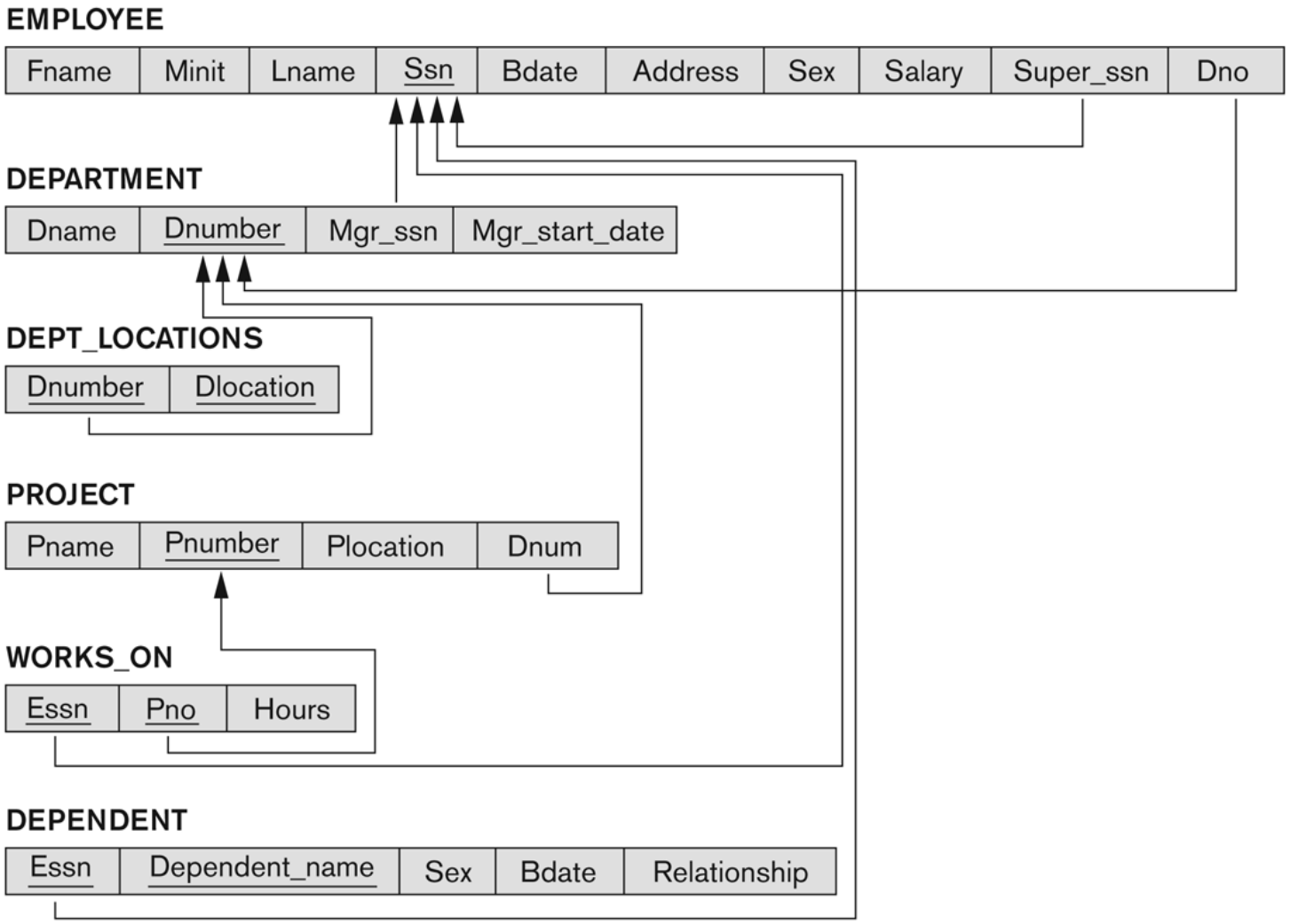
Relational Algebra Overview

Relational Algebra consists of several groups of operations

- Unary Relational Operations
  - SELECT (symbol: **σ** (sigma))
  - PROJECT (symbol: **π** (pi))
  - RENAME (symbol: **ρ** (rho))
- Relational Algebra Operations From Set Theory
  - UNION (∪), INTERSECTION (∩), DIFFERENCE (or MINUS, **–**)
  - CARTESIAN PRODUCT (**x**)
- Binary Relational Operations
  - JOIN (several variations of JOIN exist)
  - DIVISION
- Additional Relational Operations
  - OUTER JOINS, OUTER UNION
  - AGGREGATE FUNCTIONS (These compute summary of information: for example, SUM, COUNT, AVG, MIN, MAX)

Database State for COMPANY

Figure 5.7
Referential integrity constraints displayed on the COMPANY relational database schema.



Unary Relational Operations: SELECT

– In general, the *select* operation is denoted by

$\sigma_{\text{<selection condition>}}(R)$ where

- the symbol σ (sigma) is used to denote the *select* operator
- the selection condition is a Boolean (conditional) expression specified on the attributes of relation R
- tuples that make the condition **true** are selected
  – appear in the result of the operation
- tuples that make the condition **false** are filtered out
  – discarded from the result of the operation

Unary Relational Operations: SELECT

The SELECT operation (denoted by σ (sigma)) is used to select a *subset* of the tuples from a relati on based on a **selection condition**

Examples:

- Select the EMPLOYEE tuples whose department number is 4:

$$\sigma_{DNO = 4} (EMPLOYEE)$$

- Select the employee tuples whose salary is greater than $30,000:

$$\sigma_{SALARY > 30,000} (EMPLOYEE)$$

Unary Relational Operations: PROJECT (cont.)

The general form of the *project* operation is:

$$\pi_{<\text{attribute list}>}(R)$$

- π (pi) is the symbol used to represent the *project* operation
- <attribute list> is the desired list of attributes from relation R.

The project operation *removes any duplicate tuples*

- This is because the result of the *project* operation must be a *set of tuples*
  - Mathematical sets *do not allow* duplicate elements.

- Retrieve the first name, last name and salary for all employee

$\pi_{lastname,firstname,salary}$ (employee)

- retrieve the first name, last name, and salary of all employees wh o work in department number 5

$\pi_{FNAME, LNAME, SALARY}(\sigma_{DNO=5}(EMPLOYEE))$

- Retrieve data of all employee whose work in Department no 4 an d their salary grater than or equal 25000

$\sigma_{DNO=4 \text{ and salary } >=25000}$ (EMPLOYEE))

Unary Relational Operations: RENAME

The general RENAME operation ρ can be expressed by any of the following forms:

- $\rho_{S\ (B1,\ B2,\ \ldots,\ Bn\ )}(R)$ changes both:
  - the relation name to S, *and*
  - the column (attribute) names to B1, B1, ...Bn
- $\rho_S(R)$ changes:
  - the *relation name* only to S
- $\rho_{(B1,\ B2,\ \ldots,\ Bn\ )}(R)$ changes:
  - the *column (attribute) names* only to B1, B1, ...Bn

Relational Algebra Operations from Set Theory: UNION

UNION Operation

- Binary operation, denoted by ∪
- The result of R ∪ S, is a relation that includes all tuples that are either in R or in S or in both R and S
- Duplicate tuples are eliminated
- The two operand relations R and S must be “type compatible” (or UNIO N compatible)
  - R and S must have same number of attributes
  - Each pair of corresponding attributes must be type compatible (have same or compatible domains)

Relational Algebra Operations from Set Theory: UNION

Example:

- To retrieve the social security numbers of all employees who either *work in department 5* or *directly supervise an employee who works in department 5*

- We can use the UNION operation as follows:

$$RESULT \leftarrow \pi_{SSN}(\sigma_{DNO=5}(EMPLOYEE)) \cup \pi_{SUPERSSN}(\sigma_{DNO=5}(EMPLOYEE))$$


THANK
YOU


Any
Question?